Multilevel Progressive Network With Nonlocal Channel Attention for Hyperspectral Image Super-Resolution

Jianwen Hu, *Member, IEEE*, Yaoting Liu, Xudong Kang, *Senior Member, IEEE*, and Shaosheng Fan

***Abstract*— Deep convolutional neural networks (CNNs) have made great progress in the super-resolution (SR) of hyperspectral images (HSIs). However, most methods utilize convolution to explore local features, and global features are ignored. It is expected that combining nonlocal mechanism with CNN will improve the performance of HSI SR. This article presents a multilevel progressive HSI SR network. The dense nonlocal and local block (DNLB) is constructed to combine local and global features, which are used to reconstruct SR images at each level. Due to the high dimension of HSI, original nonlocal methods produce memory-expensive attention maps. We develop a nonlocal channel attention block to extract the global features of HSIs efficiently. Spatial-spectral gradient is injected in the nonlocal attention block to obtain better details. Furthermore, the progressive learning mode-based multilevel network is proposed to reconstruct HSI with fine details. A number of experiments demonstrate that our method can reconstruct HSIs more accurately than existing methods.**

***Index Terms*— Convolutional neural network (CNN), hyperspectral image (HSI) super-resolution (SR), multilevel progressive network (MPNet), nonlocal channel attention.**

I. Introduction

HYPERSPECTRAL imaging is the process of acquiring narrow and continuous images by imaging spectrometers that operate in the ultraviolet, visible, near-infrared, and mid-infrared regions of the electromagnetic spectra. In hyperspectral images (HSIs), spectral information is rich, and each pixel represents a smooth and complete spectral curve. HSIs are widely used in the detection of internal and external quality of agricultural products [1], mineral exploration [2], vegetation detection, [3], and other fields. However, due to the hardware limitations of hyperspectral imaging sensors, the spatial resolution of HSIs is lower than that of natural images. This affects the real applications such as HSI classification [4]. To solve this problem, HSI super-resolution (SR) is proposed to reconstruct high-resolution HSIs from low-resolution (LR) HSIs [5], [6].

Compared with red, green, and blue (RGB) images, HSI contain a number of bands that have spectral correlation. Hence, SR of HSIs is more difficult than that of natural images (RGB images). Many SR methods fuse high-resolution auxiliary images (Panchromatic images, RGB images, and multispectral images) and low-resolution HSIs to obtain high-resolution HSIs. HSI fusion can be divided into the following categories: 1) Bayesian-based approaches [7]; 2) tensor-based approaches [8]; 3) matrix factorization-based approaches [9]; and 4) deep learning-based approaches [10]. Deep learning-based methods can be further categorized into unsupervised [11] and supervised [12] methods. The advantage of the unsupervised methods is that they do not require a large number of labeled training samples. An unsupervised fusion method called HyCoNet [11] consists of three coupled autoencoder networks, and the point spread function (PSF) and spectral response function (SRF) are learned adaptively. Qu et al. [13] proposed an unregistered and unsupervised mutual Dirichlet-Net that fuse HSIs and multispectral images without multimodality registration. A model-inspired network [14] was designed in an unsupervised manner that involves an implicit autoencoder network. Because these methods do not learn high-resolution images, their performance needs to be strengthened. For better reconstructing the high-frequency details, a supervised method based on the edge-conditioned feature transform network [15] is proposed. However, in practice, these fusion methods are limited by the absence of auxiliary images. Auxiliary images are often difficult to obtain in practical applications. In addition, HSI and auxiliary images are acquired by different imaging sensors, which increase the hardware cost. Therefore, we will focus on the method of single HSI SR, which is more flexible and does not require additional images.

The single HSI SR reconstructs high-resolution (HR) images from single LR images. Traditional single image SR methods mainly include two categories. The first one is interpolation. The methods based on interpolation are simple and fast, but the reconstructed images are usually blurred. The second one is to utilize effective spatial or spectral statistical distribution as a priori, such as total variation [16], sparse regularization [17], low rank and self-similarity [18] to describe edges and textures. However, these methods require a priori to obtain better performance, and effective priors are usually difficult to obtain. In recent years, deep convolutional neural networks (CNNs) have been widely used in

Manuscript received 16 June 2022; revised 29 September 2022 and 27 October 2022; accepted 8 November 2022. Date of publication 10 November 2022; date of current version 28 November 2022. This work was supported in part by the National Key Research and Development Program of China under Grant 2021YFA0715203; in part by the National Natural Science Foundation of China under Grant 62271087; and in part by the Natural Science Foundation of Hunan Province, China under Grant 2021JJ40609. *(Corresponding author: Jianwen Hu.)*

Jianwen Hu, Yaoting Liu, and Shaosheng Fan are with the School of Electrical and Information Engineering, Changsha University of Science and Technology, Changsha 410114, China (e-mail: hujianwen1@163.com; 1486296559@qq.com; fss508@163.com).

Xudong Kang is with the College of Robotics, Hunan University, Changsha 410082, China (e-mail: xudong_kang@163.com).

Digital Object Identifier 10.1109/TGRS.2022.3221550

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various image-processing fields and achieved amazing results. Some SR methods based on CNN have been proposed. The first SR network is super-resolution convolutional neural network (SRCNN) [19], which proves that CNN has more advantages than traditional methods in image SR and can perform end-to-end mapping. The SR models are then improved by increasing the network depth, which can improve the nonlinearity of network models, thus allowing more complex modeling [20]. To improve the long-range modeling capability of the network, Mei et al. [21] proposed a novel nonlocal sparse attention network (NLSN) with dynamic sparse attention pattern for SR of natural images.

As CNNs perform satisfactorily in natural image SR, they are also applied to HSIs [22], [23], [24], [25]. These methods use 2-D CNN to implement SR. Although 2-D convolution is helpful in spatial resolution, they fail to consider the spectral correlation among adjacent bands of HSIs, resulting in spectral distortion. Compared with 2-D CNN, 3-D CNN can extract both spectral and spatial features. Thus, 3-D CNN can improve the spatial resolution of HSIs while protecting the spectral correlation. Recently, some HSI SR methods based on 3-D CNN have been proposed. Mei et al. [26] first proposed HSI SR based on 3-D full convolutional neural network (3D-FCNN). A multiscale wavelet 3-D CNN was designed by Yang et al. [27]. Li et al. [28] proposed 3-D attention-based generative adversarial network for HSI SR. However, 3-D convolution (3-D CONV) significantly increases the number of parameters and the amount of computation. A 1D-2D attentional convolution neural network [29] is proposed for SR. It extracts spectral and spatial features using 1-D and 2-D convolution. Although this method reduces the number of parameters, it cannot explore spectral-spatial information simultaneously. Therefore, Wang et al. [30] proposed mixed 2-D/3-D CNN (MCNet) for HSI SR, and a novel HSI SR method by exploring the relationship between 2-D and 3-D CONV [31] is introduced. To effectively extract structural spatial-spectral correlation, a bidirectional 3-D quasi-recurrent neural network (BIQRNN) [32] was designed.

However, these methods based on CNN only considered the local feature and failed to capture the global information. Recently, various transformers [33] and nonlocal network [34] have shown remarkable potential in computer vision, which can capture long-range pixel interactions of images [35], [36]. The key of transformer is self-attention. Although nonlocal and self-attention mechanisms extract global features effectively, their complexity grows quadratically with the size of image. Due to the 3-D characteristics of hyperspectral data, more memory is required for HSI SR. Cai et al. [37] introduced the transformer to HSI compressive sensing that calculates self-attention along the spectral dimension. Liu et al. [38] proposed a separable self-attention to extract global spatial features. Nevertheless, high-frequency details of HSIs are not well exploited in these methods. Yang et al. [39] proposed a hybrid local and nonlocal 3-D network (HLNnet) for HSI SR through using a simplified nonlocal module [40], but the ability of global modeling needs to be strengthened and the number of parameters is large. And the methods of local and global feature extraction for HSI SR need to be further explored. The output feature maps of the convolutional layer often contain much redundancy [41], which is not well exploited in the process of HSI SR. In addition, the performance of a SR network is closely related to the complexity and depth of the network. To improve SR performance, neural networks are becoming deeper and deeper [42]. The vanishing gradient problem is encountered when training deep neural networks.

To solve these problems, we propose a multilevel progressive HSI SR network called multilevel progressive network (MPNet). At each level, the dense nonlocal and local block (DNLB) is proposed to extract local and global features effectively and is applicable to HSI cubes. Specifically, we design a nonlocal channel attention block with spatial-spectral gradients that generates attention maps with cheap memory and enhances reconstruction quality of high-frequency details. Because the channels of 3-D network of HSIs contain spatial and spectral features, the nonlocal channel attention block computes the attention map of channel dimension rather than spatial dimension. This manner leverages abundant spatial-spectral information to obtain better HSIs while implicitly capturing the long-range dependencies with small attention maps. To reconstruct more accurate spatial details, we propose to inject spatial-spectral details into the nonlocal channel attention block. The 3-D Ghost module is applied to embrace the redundancy in feature maps, which can extract local features effectively. In addition, the multilevel network exploits the predictions of different levels to progressively refine the HSI SR process. This progressive learning mode is capable of reconstructing HSI with fine details. Multilevel network supervised at each level can force middle layers to learn more useful features, and the vanishing gradient problem can be mitigated. In summary, the main contributions in this article can be summarized as follows.

1) We design a nonlocal channel attention block with spatial-spectral gradients to address the problem of the huge complexity in the original nonlocal module while paying more attention to high-frequency details. It captures complex mappings of spatial-spectral information while extracting edge features of HSIs sufficiently and obtains long-range dependencies with linear complexity.
2) We propose DNLB consisting of 3-D Ghost and nonlocal channel attention modules. The local and global features of HSIs are extracted simultaneously for better modeling the process of HSI SR.
3) A progressive learning mode-based multilevel network is proposed to generate more accurate SR images. The framework decomposes the challenging HSI SR task into different levels to reconstruct SR images progressively.

The rest of this article is organized as follows: First of all, Section II describes the proposed method in detail. Then Section III gives the ablation experiments of the proposed method and the comparison experiments with other methods. Finally, a conclusion is drawn in Section IV.

II. Methodology

A. Overall Network Structure

In this section, we will describe the overall structure of the proposed MPNet. As shown in Fig. 1, the MPNet is divided into N levels to progressively reconstruct SR images. Each level consists of downsampling, upsampling, and DNLB. This is a multilevel up-and-down sampling architecture, which facilitates more complex modeling. In the training process, the

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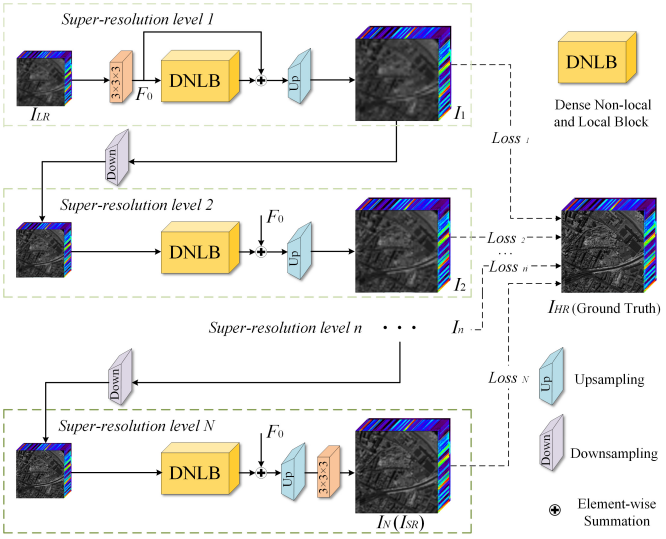


Fig. 1. Overall architecture of the proposed MPNet. Each level is composed of DNLB, upsampling and downsampling. At each level, a supervised loss with the reference image is added to optimize SR network.

images generated by each level is supervised with the reference images. Instead of simply cascading networks, we incorporate a loss at each level, which achieves multilevel supervision. In the testing possesses, reconstructed images of different levels can be obtained. The reconstructed image at the final level is used as the final SR image, which presents the best performance. Our network can alleviate the problem of vanishing gradient through multilevel loss.

Let $I_{\mathrm{LR}} \in \mathbb{R}^{(H/r)\times(W/r)\times B\times 1}$ and $I_{\mathrm{SR}} \in \mathbb{R}^{H\times W\times B\times 1}$ denote the input low resolution HSI and the output SR HSI, where W and H are the width and height of each band, and B represents the number of the bands in HSI, r represents the scale factor, and 1 is the number of channels in the feature map. First of all, a 3-D CONV is used to extract features about I_{LR}

$$F_0 = \mathrm{Conv}_{3\times3\times3}(I_{\mathrm{LR}}) \quad \left(F_0 \in \mathbb{R}^{H\times W\times B\times C}\right) \tag{1}$$

where $\mathrm{Conv}_{3\times3\times3}(\cdot)$ is 3-D CONV with the kernel $3\times3\times3$. Then F_0 will proceed to N DNLBs, and the feature map generated after each block will be upsampled to get the reconstructed images $I_1, I_2, \ldots, I_N$

$$\begin{cases} I_1 = f_{\mathrm{up}}(F_0) + f_{\mathrm{DNLB1}}(F_0) \\ I_n = f_{\mathrm{up}}(F_0) + f_{\mathrm{DNLBn}}(f_{\mathrm{down}}(I_{n-1})) \quad (n = 2, \ldots, N) \end{cases} \tag{2}$$

where $f_{\mathrm{DNLB}}(\cdot)$ represents the DNLB function, $f_{\mathrm{up}}(\cdot)$ is the deconvolutional upsampling function, and $f_{\mathrm{down}}(\cdot)$ is 3-D convolutional downsampling function.

In the training stage, the MPNet is optimized by minimizing the loss between the reconstructed HSIs $I_1, I_2, \ldots, I_N$ and the reference HSIs. The mean square error (MSE) is often used as a loss function of HSI SR based on CNN, but it is not enough to prevent the spectral distortion. Therefore, we use two loss functions, including the MSE loss and spectral gradient loss [43]. The MSE is used to calculate the square sum of the error between the SR image and the reference image. The spectral gradient loss is used to protect the spectral correlation. The MSE loss function (L_{MSE}) and spectral gradient loss function (L_G) are formulated as, equation (3), shown at the bottom of the page, where I_{HR} and I_n denote the reference image and the SR image at the nth level, respectively. I^G_{HR} and I^G_n denote the spectral gradient of the reference image and the SR image at the nth level. The loss function for each level is $\mathrm{Loss}_1, \mathrm{Loss}_2, \ldots, \mathrm{Loss}_N$. As a result, the network is more accurate and efficient as it is supervised from different levels. The loss of the nth level is expressed as follows:

$$\mathrm{Loss}_n = L_{\mathrm{MSE}}(I_{\mathrm{HR}}, I_n) + L_G(I_{\mathrm{HR}}, I_n) \quad (n = 1, \ldots, N). \tag{4}$$

Instead of directly predicting a reconstructed image, the proposed model predicts images from different levels, and progressively achieves it. The MPNet is optimized with the following loss function:

$$\mathrm{Loss}_{\mathrm{total}} = \sum_{n=1}^{N} \mathrm{Loss}_n \tag{5}$$

where $\mathrm{Loss}_{\mathrm{total}}$ represents the total loss and is the sum of the losses of all levels. The SR images of each level are supervised, which helps us to find the better network parameters.

B. Dense Nonlocal and Local Block

As shown in Fig. 2, the proposed DNLB connects three ghost module and nonlocal channel attention blocks (GNCBs) densely [44] followed by two asymmetric convolutions. GNCB can effectively extract global and local features through nonlocal channel attention and 3-D Ghost modules. The dense skip connections in DNLB enable enhanced feature propagation, while the concatenation of subblocks of features from different layers has a better effect on image reconstruction. The DNLB stacks the same subblocks for learning hierarchical spatial-spectral features. Dense connections enhance the utilization of feature maps through feature reuse, eliminate the disappearance of gradients, and enhance feature extraction. In this way, global and local features can be fully used. The lth layer receives the feature maps of all previous layers, $X_0, X_1, \ldots, X_{l-1}$, as input

$$X_l = f_{\mathrm{GNCB}}\left(\left[X_0, X_1, \ldots, X_{l-1}\right]\right). \tag{6}$$

The convolutional operation in the previous method can only extract the local features. Therefore, it is important to add nonlocal feature extraction operators. GNCB combines

$$\begin{cases} L_{\mathrm{MSE}}(I_{\mathrm{HR}}, I_n) = \dfrac{1}{HWB}\displaystyle\sum_{i=1}^{H}\sum_{j=1}^{W}\sum_{k=1}^{B}\left|I_{\mathrm{HR}}(i,j,k) - I_n(i,j,k)\right|^2 \\ L_G(I_{\mathrm{HR}}, I_n) \quad = \dfrac{1}{HWB}\displaystyle\sum_{i=1}^{H}\sum_{j=1}^{W}\sum_{k=1}^{B}\left|I^G_{\mathrm{HR}}(i,j,k) - I^G_n(i,j,k)\right| \end{cases} \tag{3}$$

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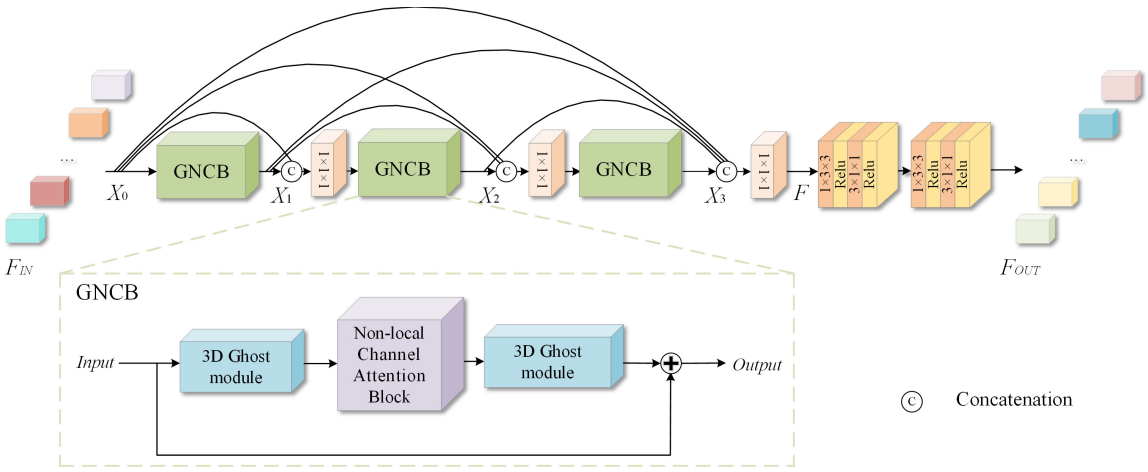


Fig. 2. DNLB used at each level of the MPNet and its subblock GNCB.

the local and nonlocal features. This module consists of three parts connected by residual: two 3-D Ghost modules and a nonlocal channel attention block.

After dense connection, we use two layers of asymmetric convolution [45] to enhance the extraction of spatial-spectral features. The $1 \times 3 \times 3$ and $3 \times 1 \times 1$ convolutions are used instead of a $3 \times 3 \times 3$ convolution, which greatly reduces the number of convolutional parameters and improves the performance of convolution feature extraction.

C. 3-D Ghost Module

Due to the existence of spectral dimensions of HSIs, 3-D CONV [see Fig. 3(a)] is used to protect the spectral correlation of HSIs. However, 3-D CONV has the problems of a lot of parameters and computation. The output feature maps of the convolutional layer usually contain a lot of redundancy [41]. Therefore, we use the 3-D Ghost module [41] instead of ordinary convolution to utilize the redundancy in feature maps and improve efficiency. Fig. 3(b) shows the 3-D Ghost module. Specifically, the 3-D Ghost module first reduces the total number of feature maps by $1 \times 1 \times 1$ convolution to obtain the intrinsic feature maps. Then, the depthwise convolution with the characteristic of cheap operation is used to augment features. The identity mapping is paralleled with depthwise convolution in the 3-D Ghost module to preserve the intrinsic features.

The depthwise convolution $\Phi(\cdot)$ in Fig. 3(b) can extract the feature maps in an efficient way, because each convolution kernel only operates on a channel. Depthwise convolution decreases the complexity of convolution computation and the number of parameters. The original Ghost module (OGM) groups each channel when performing depthwise convolution. Although the theoretical acceleration ratio is greatly improved, the hardware operation of a large number of depthwise convolutions takes a long time. This is because depthwise convolution is not suitable on some hardware platforms due to its high memory access and low arithmetic intensity. Therefore, the number of groups in the channel dimension is reduced to two groups to improve efficiency. After reducing the numbers of groups, the performance is slightly improved. It is beneficial to extract local features more efficiently and accurately.

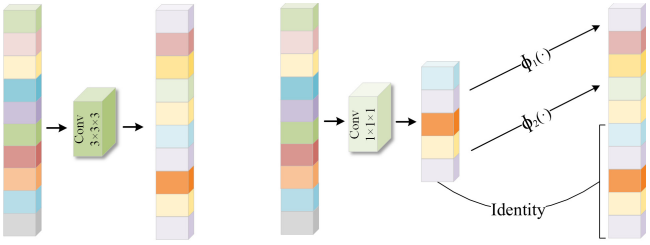


Fig. 3. Comparison between 3-D CONV and 3-D Ghost module. (a) 3-D CONV. (b) 3-D Ghost module.

D. Nonlocal Channel Attention Block With Spatial-Spectral Gradient

Previous nonlocal-based methods usually generate huge attention maps accompanied by quadratic complexity. Due to the high dimension of hyperspectral data cubes, it generates the $HWB \times HWB$ matrix during dot product, which leads to expensive memory requirements. Our nonlocal channel attention block generates small attention maps with a size of $C \times C$. Each channel can be regarded as a response of spatial-spectral information. So the nonlocal channel attention mechanism is more applicable for hyperspectral data. The complex mapping among feature maps can be enhanced by exploiting the association among channels and modeling long-range interactions implicitly. Meanwhile, high-frequency details are important in image SR. Therefore, we build a nonlocal channel attention block with spatial-spectral gradient to explicitly model interdependencies among channels while enhancing the high-frequency details of HSIs. It does not generate memory-expensive attention maps. The integration of high-frequency details with global features can help the network to reconstruct HSIs with higher quality. Fig. 4 gives the nonlocal channel attention block with spatial-spectral gradient.

The spatial-spectral edge information I_G is extracted by gradient operation. We use subtraction pixel-by-pixel to obtain the gradient map as follows:

$$\begin{aligned} I_G(i,j,k) = [&|I(i,j,k) - I(i-1,j,k)| \\ &+ |I(i,j,k) - I(i,j-1,k)| \\ &+ |I(i,j,k) - I(i,j,k-1)|]/3 \end{aligned} \tag{7}$$

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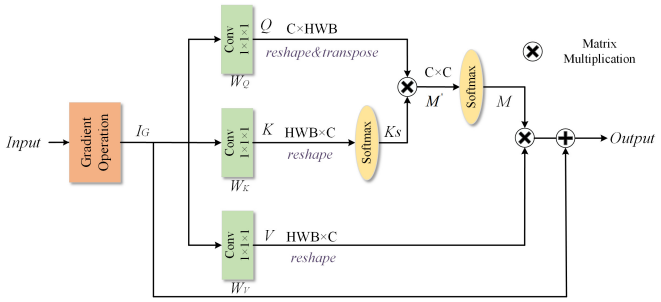


Fig. 4. Nonlocal channel attention block with spatial-spectral gradient.

In nonlocal channel attention block, first, we get Q, K, and V from the spatial-spectral image edge by the $1 \times 1 \times 1$ convolutional operation of $W_Q(\cdot)$, $W_K(\cdot)$, and $W_V(\cdot)$

$$Q = W_Q(I_G), \quad K = W_K(I_G), \quad V = W_V(I_G)$$
$$Q,\ K,\quad V \in \mathbb{R}^{H\times W\times B\times C}. \tag{8}$$

Then the feature matrix K is transformed into $\mathbb{R}^{HWB\times C}$, and the softmax operation is applied to the columns of K to obtain K_S. K_S is multiplied by the vector obtained by transposing Q to get $M' \in \mathbb{R}^{C\times C}$. And the softmax layer is applied to the columns of M' to obtain the global channel attention map M

$$M_{ij} = \frac{\exp\left(Q_i \times K_{S_j}\right)}{\sum_{i=1}^{C} \exp\left(Q_i \times K_{S_j}\right)} \tag{9}$$

where M_{ij} measures the impact of the ith channel on the jth channel. M facilitates information interaction among channels. The attention map M is a matrix of comparatively small dimension due to $C \ll$ HWB. Thus, our nonlocal attention mechanism avoids expensive attention maps. Fu et al. proposed a nonlocal channel attention module in [46], but this module does not fully exploit the information among channels. In our nonlocal module, the softmax layer is used to enhance the interchannel information interaction before generating the global channel attention map, and achieve better performance. It enables content interaction among channels without generating expensive attention maps. Moreover, nonlocal channel attention block with spatial-spectral gradient can exploit the global high-frequency details of the feature maps, which is conducive to restoring spatial edges.

The formula of the nonlocal channel attention block with spatial-spectral gradient operation can be expressed as

$$\text{output} = f_{\text{NCAB}}(I_G) + I_G. \tag{10}$$

III. Experiments

In this section, the performance of our method is evaluated both qualitatively and quantitatively. First, the dataset, evaluation indexes, and implementation details are given. Then, the ablation study is performed to demonstrate the effectiveness of our method. Finally, the proposed method is compared with other hyperspectral SR methods.

A. Experimental Details

1) Datasets: To conduct a comprehensive evaluation of the proposed MPNet, three HSI datasets are employed for the experiment, including the Pavia Center, Chikusei, and the Houston datasets.

Pavia Center Dataset [47]: The size of the Pavia Center image is $1096 \times 715 \times 102$. The height and width are 1096 and 715, respectively, and there are 102 bands, with wavelengths ranging from 430 to 830 nm. We remove two noisy bands and divide the image into 30% test region ($1096 \times 200 \times 100$) and 70% training region ($1096 \times 515 \times 100$). The training data are rotated by three angles and cut into 8790 samples, and the sample size is $36 \times 36 \times 100$. The test dataset consists of seven images with the size of $144 \times 144 \times 100$.

Chikusei Dataset [48]: The Chikusei dataset contains 128 bands, wavelength ranging from 343 to 1018 nm, with a size of 2517×2335. There are 19 types of ground objects, including urban and rural areas, which is taken by Headwall Hyperspec-VNIR-C imaging sensor over agricultural and urban areas in Chikusei, Ibaraki, Japan. We extract eight nonoverlap regions with $204 \times 204 \times 128$ pixels to form the test data. The remaining area is used as training set, which is cut into 7790 samples of size $36 \times 36 \times 128$.

Houston Dataset [49]: The Houston dataset was released over the University of Houston campus by the Hyperspectral Image Analysis Laboratory and the National Center for Airborne Laser Mapping (NCALM), which is consists of 48 spectral bands, and its spectral range from 380 to 1050 nm. The scene contains 601×2384 pixels and is divided into a training set and a test set. The training sets are cut into 12 470 samples with the size of $36 \times 36 \times 48$. The test dataset consists of 16 images with the size of $144 \times 144 \times 48$.

To ensure a reliable evaluation for super-resolved images, we generate the LR HSIs with reduced scales and take the original HSI as reference, according to the Wald's protocol [50]. Thus, the image degradation method that is employed to produce reduced resolution images is crucial. The image degradation method must take the modulation-transfer function (MTF) of the imaging system into account. The gaussian filters can be tuned to closely match the sensor MTF [51] to simulate the image degradation. In our experiments, the simulated LR HSIs are generated by gaussian downsampling from a reference HSI with three-scale factors, i.e., 2, 3, and 4, and the SR results are evaluated on reference images. This manner is consistent with the recent HSI SR algorithms [28], [31], [39].

2) Evaluation Indexes and Implementation Details: To demonstrate the superiority of our method in terms of both spectral and spatial quality, we adopt three commonly used evaluation indexes. The mean peak signal-to-noise ratio (MPSNR) and mean structure similarity (MSSIM) [52] are used to evaluate the spatial quality, and the spectral angle mapper (SAM) [53] is used to evaluate the spectral quality. The MPSNR evaluates pixel error, and the MSSIM describes structural similarity. SAM measures the spectral similarity by the angle between the spectral vector from the reconstructed HSI and the ground-truth spectral vector. Higher MPSNR and MSSIM values mean better quality, and lower SAM values mean less spectral distortion.

The numbers of channels of convolutional layers are set to 64. Deconvolution and strided convolution are used as upsampling and downsampling, respectively. The number of DNLBs is set at 4. In each DNLB, three GNCB and two asymmetric convolutions ($1 \times 3 \times 3$ and $3 \times 1 \times 1$ convolutions) are set. In the Ghost module, the number of

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TABLE I
STUDY ON THE NUMBER OF LEVELS

	1	2	3	4	5
MPSNR	29.65	29.70	29.77	29.92	29.80
MSSIM	0.9786	0.9791	0.9794	0.9800	0.9794
SAM	2.52	2.49	2.56	2.46	2.55

TABLE II
STUDY ON SUPERVISED MODES OF THE MPNET

.	4-th	1,4-th	1,2,4-th	1,2,3,4-th	Dense supervision
MPSNR	29.83	29.80	29.87	29.92	29.82
MSSIM	0.9797	0.9794	0.9797	0.9800	0.9792
SAM	2.54	2.56	2.49	2.46	2.51

channels in feature maps decrease from 64 to 32 by $1 \times 1 \times 1$ convolution, and the number of groups in the Ghost module in the channel dimension is set to 2. The adaptive moment estimation (ADAM) [54] is used to optimize the loss of the network, and set the batchsize to 16. The weights are initialized by the Kaiming initialization. The initial learning rate is set at 0.0005 and halved every 30 epochs. We implement the proposed method on the NVIDIA GTX 1080 Ti graphics processing unit (GPU) using the Pytorch framework.

B. Ablation Study

The ablation experiments on the Houston dataset with a scale factor of 4 were performed to prove the effectiveness of our method. First of all, we set the number of levels from 1 to 5 to analyze the effect of the number of levels on the performance. Table I shows the results of 1 to 5 levels. With the increase in the number of levels, the metrics of MPSNR increase significantly when the number of levels is less than 5. The main reason is that multilevel networks support the quality of learning representations better. However, the performance of five levels is significantly reduced. This is because the deepnetworks may lead to overfitting and are difficult to train. It can be seen that four levels give the best result. Therefore, we set the number of levels to 4 in the subsequent experiments.

Then, to illustrate the advantages of multilevel supervision, we designed a comparative experiment of supervision at different levels. Table II shows the impact of different supervision modes on the performance of image reconstruction. Compared with ordinary single supervision, multilevel supervision has obvious performance advantages without increasing the computational burden. The supervision for all levels gets the best performance. We perform an experiment of dense supervision among different levels. The experimental results show that the dense supervision degrades the SR performance. It is possible that the SR result of the initial level affects the SR result of the final level, which is detrimental to the quality of reconstructed HSIs. The results show that multilevel supervision can obtain better SR images.

In the four levels network obtained from the previous experiments, we also test the reconstructed images of each level. The results are given in Table III. With the increase of levels, the evaluation metric is progressively getting better. We have reached the target of progressive SR. Meanwhile,

TABLE III
OUTPUT RESULTS FOR EACH LEVEL OF THE MPNET AND THE GCM

	1-st	2-nd	3-rd	4-th	GCM
MPSNR	29.68	29.81	29.88	29.92	29.74
MSSIM	0.9790	0.9796	0.9798	0.9800	0.9792
SAM	2.58	2.55	2.52	2.46	2.57

TABLE IV
PERFORMANCE AND THE NUMBER OF PARAMETERS OF DIFFERENT 3-D CONV TYPES

	Params(K)	MPSNR	MSSIM	SAM
3D MGM is replaced with 3D OGM	1904	29.82	0.9794	2.58
3D MGM is replaced with 3D CONV	4489	29.84	0.9798	2.51
Remove 3D MGM	1834	29.81	0.9793	2.54
AC CONV is replaced with 3D CONV	2706	29.89	0.9798	2.53
Ours	2215	29.92	0.9800	2.46

TABLE V
ABLATION STUDY ABOUT THE NONLOCAL CHANNEL ATTENTION BLOCK

	MPSNR	MSSIM	SAM
GCB	29.71	0.9794	2.47
GCCM	29.84	0.9795	2.49
Remove gradient operation	29.81	0.9793	2.48
Remove the first softmax	29.70	0.9791	2.58
Non-local module is replaced with CONV	29.82	0.9793	2.55
Remove non-local module	29.73	0.9792	2.52
Ours	29.92	0.9800	2.46

we conduct an experiment of four DNLBs in general cascade mode (GCM) without the up-and-down sampling manner of progressive learning. The result presents a drop in MPSNR from 29.92 to 29.74 dB. The progressive learning model can help the network reconstruct more accurate HSIs.

3-D CONV has been widely used in hyperspectral SR and has shown better results. However, the large number of parameters in 3-D CONV is a serious problem. We use the 3-D Ghost module and asymmetric convolution ($1 \times 1 \times 3$ and $3 \times 3 \times 1$ convolution) to reduce the number of parameters and improve the efficiency of the 3-D CNN. The OGM is not conducive to the connection among the channels when it groups each channel. In the modified 3-D ghost module (3-D MGM), the number of groups is set to 2, which can improve reconstruction accuracy with the increase of tiny parameters compared to the OGM. The experimental results and the number of parameters (Params) are shown in Table IV. We compare the 3-D MGM, the OGM (3-D OGM), and the 3-D CONV. Such as Table IV, it can be seen that the 3-D MGM shows the better experimental results. On the other hand, using asymmetric convolution (ac CONV) instead of a 3-D CONV slightly improves SR performance and reduces the number of parameters. Therefore, the experimental results show the effectiveness of the modified Ghost module and asymmetric convolution.

Finally, the advantages of nonlocal channel attention block with spatial-spectral gradient are analyzed, and the experimental results are presented in Table V. Due to the high

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TABLE VI
QUANTITATIVE COMPARISON OF ALL METHODS ON THE PAVIA CENTER DATASET. THE RED AND BLUE INDICATE THE BEST AND SECOND-BEST RESULTS, RESPECTIVELY

Scale	Metric	Bicubic	SRCNN	3D-FCNN	VDSR	NLSN	BIQRNN	HLNnet	MCNet	Ours
×2	MPSNR	31.06	33.92	33.42	34.06	34.18	34.80	35.09	35.04	35.21
	MSSIM	0.9268	0.9564	0.9592	0.9646	0.9649	0.9725	0.9748	0.9746	0.9754
	SAM	4.64	4.39	4.22	3.97	4.24	3.78	3.66	3.68	3.59
	Params(K)	-	8	39	665	1702	1889	9776	1928	1600
	Inference Time(s)	-	1.04	1.18	1.25	1.73	1.12	1.23	1.08	1.48
×3	MPSNR	28.55	29.55	29.87	29.94	30.07	30.36	30.52	30.30	30.71
	MSSIM	0.8658	0.9069	0.9117	0.9154	0.9149	0.9297	0.9323	0.9294	0.9347
	SAM	5.57	5.49	5.21	5.46	5.62	5.08	5.15	5.33	4.89
	Params(K)	-	8	39	665	1888	1889	9831	2039	1877
	Inference Time(s)	-	1.07	1.16	1.28	1.48	1.13	0.94	0.88	1.10
×4	MPSNR	26.89	27.53	27.89	27.95	27.98	28.35	28.39	28.41	28.64
	MSSIM	0.7927	0.8477	0.8584	0.8634	0.8584	0.8873	0.8872	0.8884	0.8921
	SAM	6.30	6.44	5.95	6.41	6.55	5.89	6.11	6.08	5.63
	Params(K)	-	8	39	665	1852	1889	9899	2174	2215
	Inference Time(s)	-	1.06	1.14	1.27	1.46	1.13	0.87	0.79	0.98

dimension of 3-D hyperspectral data, the original nonlocal module is not be applied. Therefore, our nonlocal channel attention block is compared with other attention modules that have been proven to be effective. These blocks include the global context block (GCB) [40] and the global channel context module (GCCM) [46]. These blocks of global feature extraction also use 3-D mode. From Table V, it can be seen that our nonlocal channel attention algorithm achieves better results in three evaluation metrics than other attention methods. Several core components of the nonlocal channel attention block are also examined. The combination of spatial-spectral gradient and nonlocal attention mechanism can fully extract the edge details of HSIs. When spatial-spectral gradient operations are removed, MPSNR decreases by 0.11. In addition, we apply a softmax layer to the attention map in advance and further improve the performance. The MPSNR of the proposed method is 0.22 dB larger than that of removing the first softmax. To further verify the importance of global information, the nonlocal block is replaced with convolution (CONV) in the experiments. It is found that obtaining global information by nonlocal block can improve the performance of the model. The MPSNR decreases by 0.19 dB when the nonlocal module is removed, which justifies the significance of global features. As shown in Table IV, the MPSNR decreases by 0.11 dB when the 3-D Ghost is removed. The two experiments prove that better SR images can be obtained by integrating local and global features.

C. Simulated HSI SR Experiments

The proposed method is a supervised HSI SR algorithm, while unsupervised methods usually suffer from poor performance [55]. Therefore, the compared SR methods based on deep learning are supervised. Specifically, the proposed method is compared with eight methods and can be classified into three categories, i.e., interpolation method, natural image SR methods, and HSI SR methods. First, bicubic is a classic interpolation method; Second, SRCNN [19], very deep convolutional network (VDSR) [20], and NLSN [21] are the classic and recent natural image SR methods. These methods are representative of networks with different depth (i.e., SRCNN and VDSR) and nonlocal network (i.e., NLSN). Third, 3D-FCNN [26], BIQRNN [32], HLNnet [39], and MCNet [30] are the classic and recent HSI SR methods. These methods are representative of presampling methods (i.e., 3D-FCNN and BIQRNN), local and global feature extraction network (i.e., HLNnet), and mixed 2-D/3-D convolutional network (i.e., MCNet). Three datasets, i.e., Pavia Center, Chikusei, and Houston, are used to verify the superiority of the proposed method.

1) Pavia Center Dataset: As shown in Table VI, our MPNet gives the best MPSNR, MSSIM, and SAM. Specifically, the bicubic method is the worst in the all methods. The 3D-FCNN presents higher MPSNR and lower SAM than the SRCNN and interpolation methods. This is attributed to the 3-D CONV used by 3D-FCNN, which can extract spatial and spectral features simultaneously. However, 3D-FCNN is limited by the number of network layers (only three). VDSR and NLSN obtained better results through deeper networks. MCNet achieves a higher MPSNR (+0.43) than NLSN with a scale factor of 4. This is because NLSN is a natural image-based method that does not consider spectral correlation of HSI. The MPSNR of our method is 0.19 dB higher than that of HLNnet with a scale of 3, while it is 0.23 dB higher than that of MCNet with a scale of 4. Therefore, the proposed method is more effective at local and global feature extraction and presents better performance of image SR. From Table VI, we can see that the 3-D CNN is better than the 2-D CNN, and the networks with the nonlocal attention module are more conducive to HSI reconstruction. Our method uses 3-D network to protect spectral correlation, while using local and nonlocal modules to better achieve HSI SR.

And we give the number of parameters (Params) for all comparison methods with the three-scale factors in Table VI. It can be seen that our method almost possesses the least number of parameters among recent HSI SR methods (i.e., BIQRNN, HLNnet, and MCNet). Compared with the HLNnet, our method achieves better local and global feature extraction while significantly reducing the number of parameters. While for the preupsampling HSI SR methods (i.e., 3D-FCNN and BIQRNN), although the number of parameters is small, they are poor in SR performance. For the natural image SR methods (i.e., SRCNN, VDSR, and NLSN), HSI SR is performed in a band-by-band manner, so the spectral

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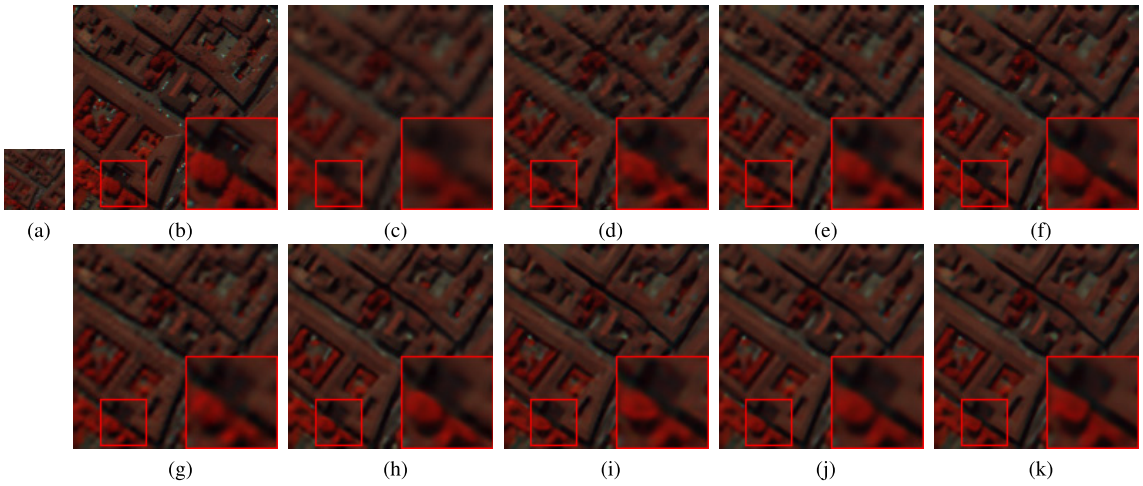


Fig. 5. HSI SR results in the Pavia Center dataset (scale factor 4). (a) LR image. (b) Reference HSI. (c) Bicubic. (d) SRCNN. (e) 3D-FCNN. (f) VDSR. (g) NLSN. (h) BIQRNN. (i) HLNnet. (j) MCNet. (k) Ours. The false color image is used for clear visualization (red: 100, green: 30, and blue: 10).

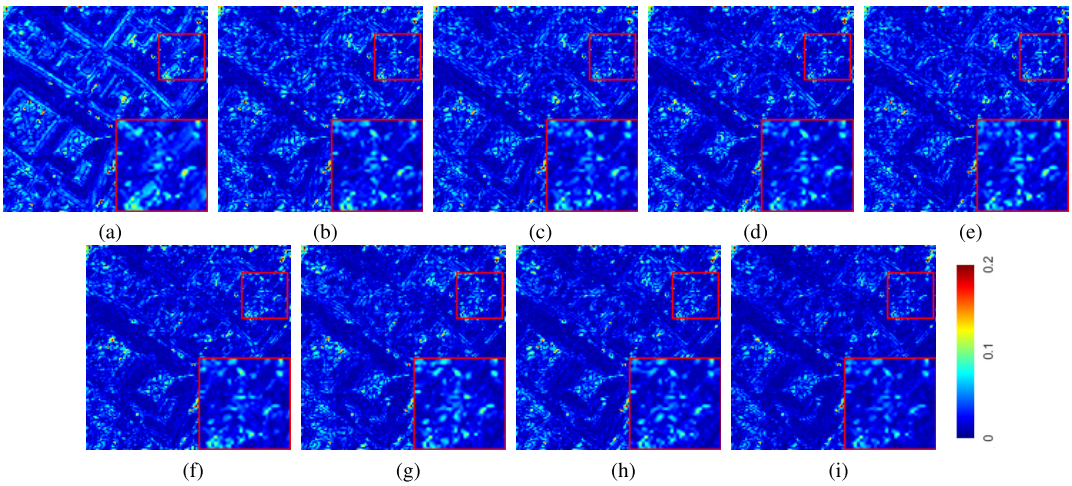


Fig. 6. Error images between the reconstructed images and the reference image in the Pavia Center dataset (scale factor 4). (a) Bicubic. (b) SRCNN. (c) 3D-FCNN. (d) VDSR. (e) NLSN. (f) BIQRNN. (g) HLNnet. (h) MCNet. (i) Ours.

correlation is protected poorly and the spatial information of adjacent bands is not be well utilized. To investigate the time consumption of the SR process, we list the average inference time on the Pavia Center dataset with three-scale factors in Table VI. The NLSN method requires more layers to obtain better results, resulting in expensive time consumption. The inference time is relatively longer for the preupsampling methods (e.g., SRCNN, 3D-FCNN, VDSR, and BIQRNN) which perform most computations in high-dimensional space. In the recent HSI SR algorithms, the proposed method can obtain better evaluation metrics with acceptable time consumption.

Fig. 5 shows the SR result of the Pavia Center test images and enlarges the same area marked by interpolation to show details. From the magnification region, it can be seen that compared with the bicubic interpolation method, the natural image SR methods based on CNN (i.e., SRCNN, VDSR, and NLSN), and the HSI SR methods based on CNN (i.e., 3D-FCNN, BIQRNN, HLNnet, and MCNet) obviously increase spatial details and show better image quality. The MPNet generates more realistic visual results and maintains the main structural information. For better evaluation of HSI reconstruction, we also provide the error images between the reference image and the reconstructed images obtained by different SR methods in Fig. 6. If the error is smaller, the reconstructed image is closer to the reference image. In general, the bluer the absolute error map is, the better the reconstructed image is. We can find that the proposed method achieves the smallest reconstructed error, while the error images of other competing methods contain more obvious red.

2) Chikusei Dataset: We conducted further experiments on Chikusei datasets to verify the SR performance. From the quantitative results in Table VII, the proposed method is better than other methods in three metrics. Compared with the recent methods (i.e., HLNnet, MCnet), our method obtains higher MPSNR and lower SAM on three scale factors. When the scale factor is 4, the MPSNR of our method is 0.17 dB higher than that of MCNet and 0.22 dB higher than that of HLNnet, while the SAM of our method is 0.15 lower than that of MCNet and 0.16 lower than that of HLNnet. This further validates that the proposed method is superior. Fig. 7 shows the reference image and the reconstructed HSIs. The results of SRCNN and 3D-FCNN are blurry. The HLNnet can restore boundary details, but still produce blurred and curved edges. In contrast, our method has a straight and

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TABLE VII

QUANTITATIVE COMPARISON OF ALL METHODS ON THE CHIKUSEI DATASET. THE RED AND BLUE INDICATE THE BEST AND SECOND-BEST RESULTS, RESPECTIVELY

Scale	Metric	Bicubic	SRCNN	3D-FCNN	VDSR	NLSN	BIQRNN	HLNnet	MCNet	Ours
×2	MPSNR	32.74	35.23	35.46	36.02	36.17	36.22	37.09	37.05	37.26
	MSSIM	0.9682	0.9824	0.9813	0.9852	0.9853	0.9853	0.9928	0.9927	0.9930
	SAM	2.12	1.68	1.71	1.68	1.63	1.60	1.44	1.48	1.38
×3	MPSNR	29.73	31.05	31.18	31.44	31.68	31.51	32.46	32.55	32.60
	MSSIM	0.9364	0.9574	0.9565	0.9593	0.9610	0.9605	0.9795	0.9797	0.9800
	SAM	2.99	2.60	2.65	2.79	2.66	2.70	2.31	2.28	2.19
×4	MPSNR	27.71	28.78	28.87	28.87	28.91	28.94	29.83	29.88	30.05
	MSSIM	0.8959	0.9254	0.9252	0.9264	0.9272	0.9543	0.9619	0.9618	0.9635
	SAM	3.84	3.38	3.37	3.73	3.66	3.46	3.07	3.06	2.91

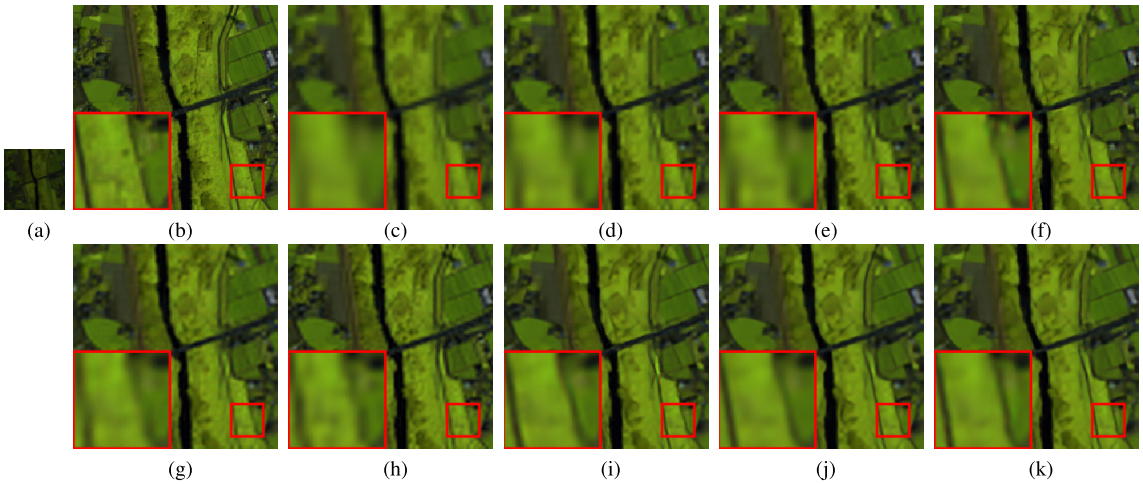


Fig. 7. HSI SR results in the Chikusei dataset (scale factor 4). (a) LR image. (b) Reference HSI. (c) Bicubic. (d) SRCNN. (e) 3D-FCNN. (f) VDSR. (g) NLSN. (h) BIQRNN. (i) HLNnet. (j) MCNet. (k) Ours. The false color image is used for clear visualization (red: 60, green: 27, and blue: 17).

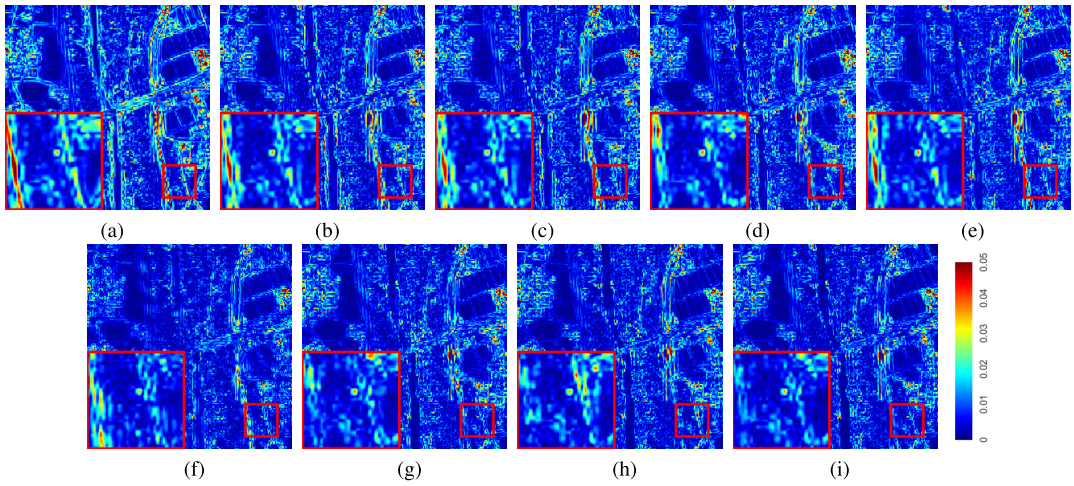


Fig. 8. Error images between the reconstructed images and the reference image in the Chikusei dataset (scale factor 4). (a) Bicubic. (b) SRCNN. (c) 3D-FCNN. (d) VDSR. (e) NLSN. (f) BIQRNN. (g) HLNnet. (h) MCNet. (i) Ours.

clear boundary, restoring almost all the road details. The error image is given in Fig. 8, which indicates that the HSI reconstructed by our method is more consistent with the reference image. This proves that the MPNet has a better performance in spatial reconstruction and generates more realistic visual results.

3) Houston Dataset: Table VIII shows the evaluation indexes of different SR algorithms for different scale factors on the Houston dataset. The evaluation indexes are averaged across 16 test images. The red and blue represent the best and second-best results, respectively. It can be observed that all the evaluation indexes of our method are the best. When

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TABLE VIII

QUANTITATIVE COMPARISON OF ALL METHODS ON THE HOUSTON DATASET. THE RED AND BLUE INDICATE THE BEST AND SECOND-BEST RESULTS, RESPECTIVELY

Scale	Metric	Bicubic	SRCNN	3D-FCNN	VDSR	NLSN	BIQRNN	HLNnet	MCNet	Ours
×2	MPSNR	32.72	35.20	35.44	35.66	35.88	36.03	36.79	36.72	37.05
	MSSIM	0.9896	0.9935	0.9942	0.9948	0.9958	0.9950	0.9959	0.9957	0.9961
	SAM	1.82	1.78	1.73	1.72	1.76	1.55	1.48	1.42	1.37
×3	MPSNR	29.31	30.18	30.23	31.62	31.25	31.69	32.09	32.55	32.79
	MSSIM	0.9770	0.9812	0.9817	0.9862	0.9870	0.9869	0.9875	0.9892	0.9897
	SAM	2.48	2.44	2.31	2.22	2.26	2.24	2.38	1.99	1.92
×4	MPSNR	26.78	28.55	28.60	28.97	29.26	29.54	29.41	29.78	29.92
	MSSIM	0.9582	0.9730	0.9732	0.9747	0.9770	0.9784	0.9781	0.9794	0.9800
	SAM	3.20	2.99	2.84	2.87	2.77	2.66	2.73	2.56	2.46

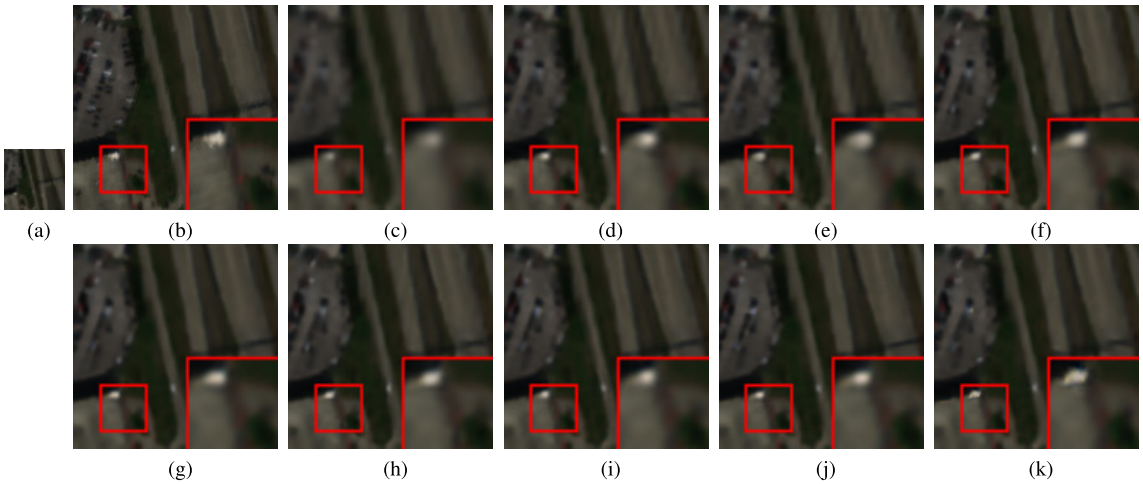


Fig. 9. HSI SR results in the Houston dataset (scale factor 4). (a) LR image. (b) Reference HSI. (c) Bicubic. (d) SRCNN. (e) 3D-FCNN. (f) VDSR. (g) NLSN. (h) BIQRNN. (i) HLNnet. (j) MCNet. (k) Ours. The false color image is used for clear visualization (red: 22, green: 14, and blue: 8).

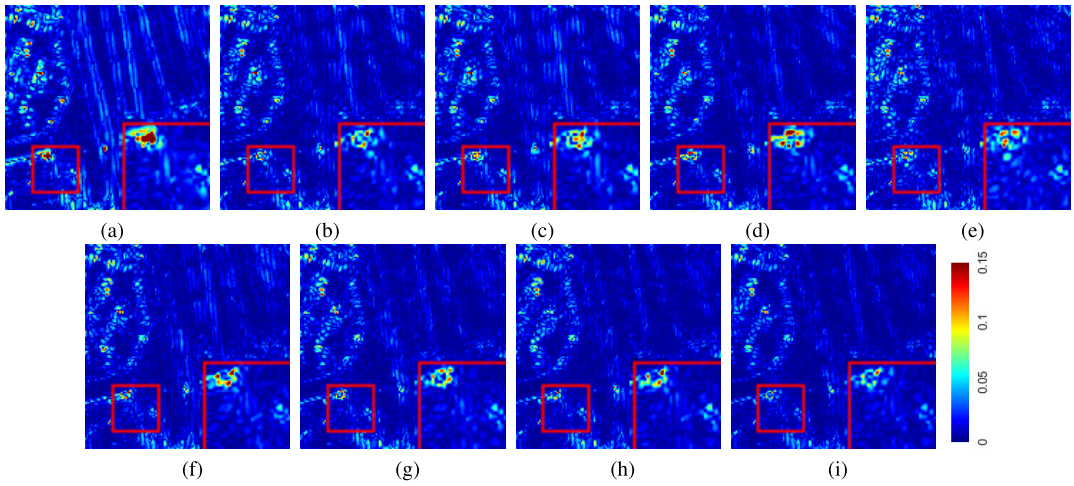


Fig. 10. Error images between the reconstructed images and the reference image in the Houston dataset (scale factor 4). (a) Bicubic. (b) SRCNN. (c) 3D-FCNN. (d) VDSR. (e) NLSN. (f) BIQRNN. (g) HLNnet. (h) MCNet. (i) Ours.

the scale factor is 2, the MPSNR of our method is 0.26 dB higher than that of HLNnet. In addition, in these methods, SRCNN, VDSR, and NLSN use 2-D convolution for HSI SR, while other deep learning methods use 3-D CONV. We can find that the overall performance of the SR network with 3-D CONV is better than the network with 2-D convolution. This is because 3-D CONV can better protect spectral correlation and improve feature extraction. Fig. 9 shows the SR images in the Houston dataset with a scale factor of 4 on different methods. The 22th, 14th, and 8th bands are used as R, G, and B bands for visualization, respectively. To better observe the effect of image reconstruction, we enlarge the red rectangular area to display more details. We observe that the SR image of MCNet is blurred in the white block. The proposed method

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TABLE IX

QUANTITATIVE COMPARISON OF ALL METHODS ON THREE DATASETS WITH A SCALE FACTOR OF 8. THE RED AND BLUE INDICATE THE BEST AND SECOND-BEST RESULTS, RESPECTIVELY

Dataset	Metric	Bicubic	SRCNN	3D-FCNN	VDSR	NLSN	BIQRNN	HLNnet	MCNet	Ours
Pavia Centre	MPSNR	23.30	23.45	23.65	23.77	23.89	24.34	24.42	24.65	24.84
	MSSIM	0.6123	0.6223	0.6888	0.6418	0.6507	0.7001	0.7026	0.7089	0.7106
	SAM	7.73	7.78	7.62	7.88	7.67	7.61	7.64	7.60	7.55
Chikusei	MPSNR	24.12	24.32	24.55	24.47	24.66	25.29	24.89	25.39	25.47
	MSSIM	0.7599	0.8702	0.8734	0.8752	0.8761	0.8864	0.8860	0.8862	0.8879
	SAM	5.68	5.86	5.32	5.66	5.61	5.37	5.58	5.33	5.19
Houston	MPSNR	22.54	22.98	23.30	23.42	23.39	23.65	23.69	23.91	24.00
	MSSIM	0.8816	0.8889	0.9073	0.9098	0.9095	0.9136	0.9173	0.9207	0.9216
	SAM	5.19	5.27	4.73	5.15	5.22	4.76	4.88	4.66	4.46

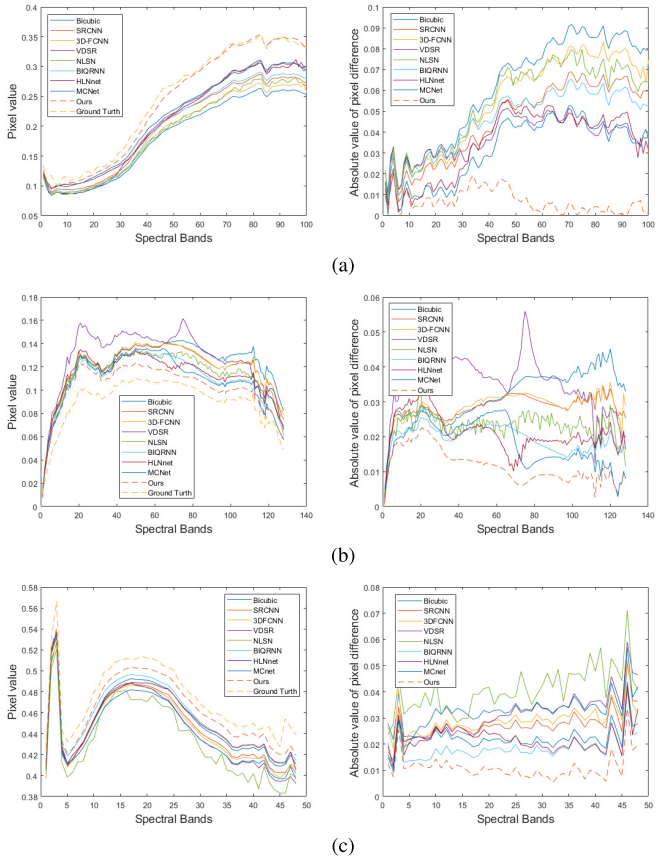


Fig. 11. Spectral curves and difference curves on a selected pixel value of three datasets with the scale factor of 4. (a) Pavia Center. (b) Chikusei. (c) Houston.

reconstructs sharper edges. It is possible that our method injects spatial-spectral details into the nonlocal channel attention block and reconstructs better edge details. To further illustrate the advantages of our method, Fig. 10 shows the error images between the reconstructed images and the reference image. Our method has the smallest error and is closer to the reference image.

To demonstrate the superiority of our method in the preservation of spectral information, we visualize the spectral vector of SR images. As shown in Fig. 11, we randomly select a pixel from each dataset with a scale factor of 4 to analyze the spectral distortion. The spectral values of the reconstructed images are presented by spectral curves. The error curves are obtained by subtracting the pixel value of the reference image from each of the reconstructed images. Then we take the absolute value. It can be seen that the reconstructed image by our method is closest to the ground truth on the Pavia Center dataset. The error curve can more intuitively compare the magnitude of spectral distortion. Ours has the smallest spectral distortion. We also compared the degree of spectral distortion on the Chikusei and Houston datasets. The spectral curves of the proposed method are closer to the ground truth, which indicates that the proposed method achieves higher performance and the spectral information is well preserved.

D. HSI SR Experiments With Large Scale Factor

To demonstrate the effectiveness of our method on a scale factor larger than 4, we conduct the HSI SR experiment with a scale factor of 8. In particular, the training sets on three datasets with a size of 32 × 32 are applied for HSI SR experiments. The training and testing regions are consistent with the previous HSI SR experiment. The evaluation metrics are given in Table IX. The red and blue indicate the best and second-best results, respectively. Our method is the best in all evaluation metrics. The MPSNR of our method is higher (+0.19, +0.18, and +0.09 dB) than that of MCNet in the Pavia Center dataset, the Chikusei dataset, and the Houston dataset. The SAM of our method is lower (−0.05, −0.14, and −0.20 dB) than that of MCNet in the Pavia Center dataset, the Chikusei dataset, and the Houston dataset. Therefore, compared with other methods, our method can reconstruct better spatial details while better preserving the spectral correlation for large-scale factor.

E. Real HSI SR Experiments

In this section, we conduct experiments on real HSI to identify the performance of the proposed method. We use the test images without any degradation operation. For the test images of the Pavia Center, Chikusei, and Houston datasets, an area with a height and width of 100 × 100 was selected. Then, all methods are applied to these areas with a scale factor of 4. The network weights trained in the simulated experiments are used in the real HSI SR experiments.

A reconstructed image with a scale factor of 4 on each test image in each dataset is shown here. To display it more clearly, the blue rectangular areas are enlarged in all the reconstructed images. Figs. 12–14 show the reconstructed

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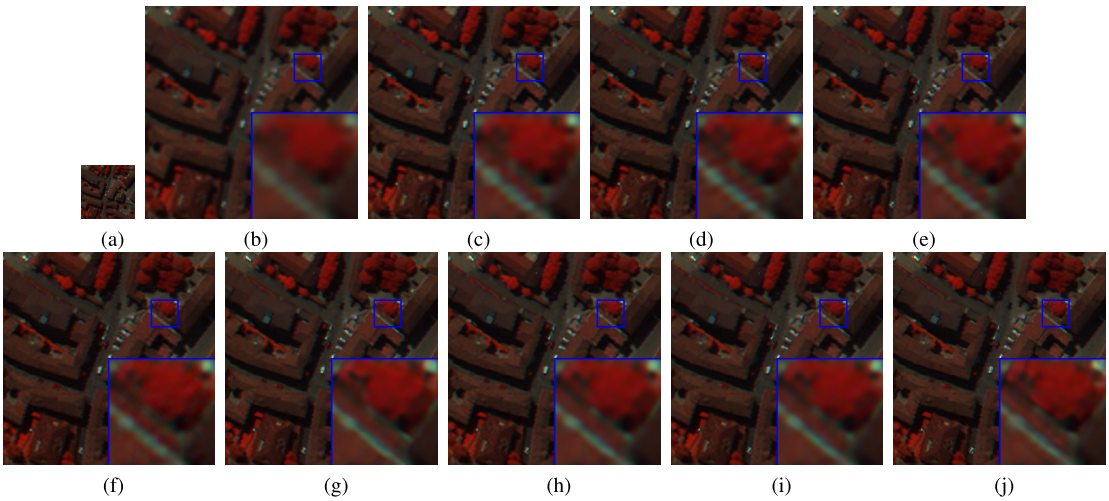


Fig. 12. Real HSI SR results in the Pavia Center dataset (scale factor 4). (a) Original HSI. (b) Bicubic. (c) SRCNN. (d) 3D-FCNN. (e) VDSR. (f) NLSN. (g) BIQRNN. (h) HLNnet. (i) MCNet. (j) Ours. The false color image is used for clear visualization (red: 100, green: 30, and blue: 10).

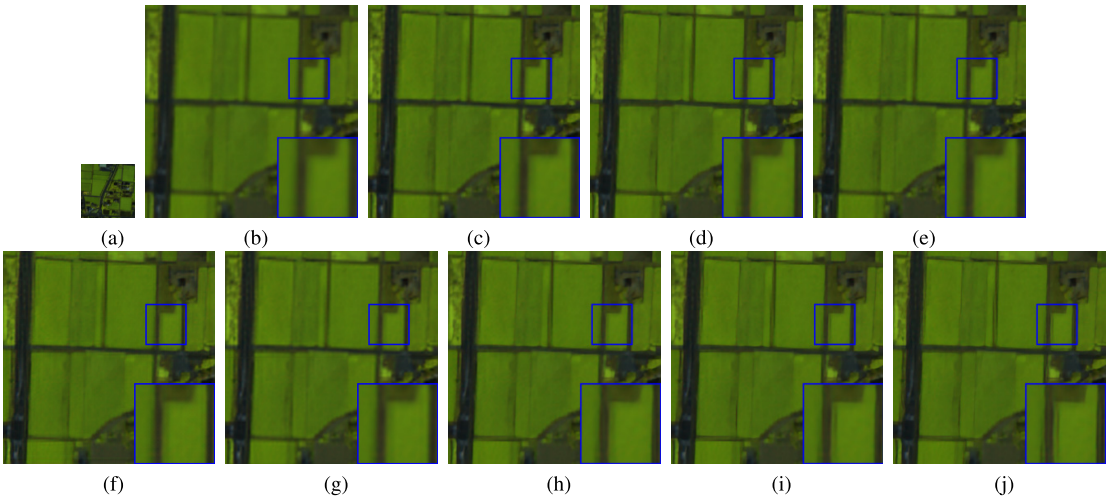


Fig. 13. Real HSI SR results in the Chikusei dataset (scale factor 4). (a) Original HSI. (b) Bicubic. (c) SRCNN. (d) 3D-FCNN. (e) VDSR. (f) NLSN. (g) BIQRNN. (h) HLNnet. (i) MCNet. (j) Ours. The false color image is used for clear visualization (red: 60, green: 27, and blue: 17).

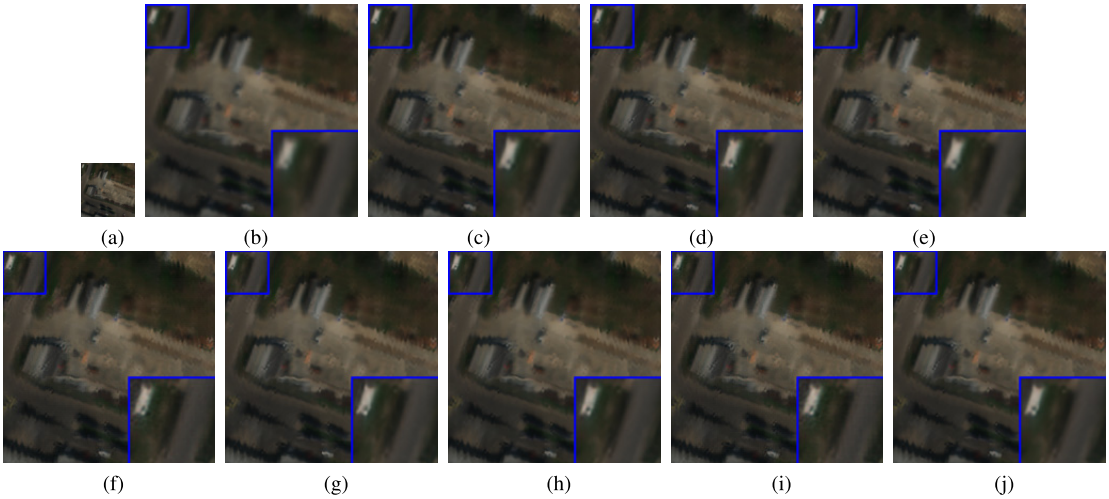


Fig. 14. Real HSI SR results in the Houston dataset (scale factor 4). (a) Original HSI. (b) Bicubic. (c) SRCNN. (d) 3D-FCNN. (e) VDSR. (f) NLSN. (g) BIQRNN. (h) HLNnet. (i) MCNet. (j) Ours. The false color image is used for clear visualization (red: 22, green: 14, and blue: 8).

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images of the Pavia Center, Chikusei, and Houston datasets, respectively. We can find that the MPNet presents the best edge details, while other competing methods lead to some distortion and blurry details. Our reconstructed image contains clearer thin lines in Fig. 12. In Fig. 13, our result presents clearer boundary details, while the edge details of other methods are blurred. As it can be seen from Fig. 14, the SR image of our method is the clearest. Therefore, our method shows superior performance under the real HSI SR.

IV. Conclusion

In this article, we propose a MPNet for hypespectral image SR that progressively improves SR images by plugging in supervision at each level. Each level can extract local and nonlocal features. The local and nonlocal parts are the 3-D GNCB with spatial-spectral gradient, respectively. The 3-D Ghost module makes local feature extraction more effective. The nonlocal channel attention block with spatial-spectral gradient is designed to extract global channel features and retain the high frequency details. The nonlocal channel attention block overcomes the problem that the original nonlocal module produces memory-expensive attention maps and is successfully applied to the HSI SR. The multisupervised training method does not add additional computational resources and improves the performance of image reconstruction. The experimental results on three HSI datasets have demonstrated that the proposed MPNet not only achieves the best performance in three metrics, but also obtains better reconstructed images in comparison with the different SR methods. In future work, we will investigate the interpretability of SR networks by combining the HSI model with CNN, and construct SR networks more effectively.

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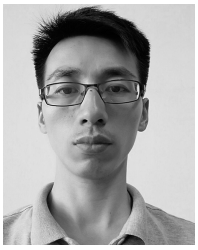
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Jianwen Hu (Member, IEEE) received the B.S. degree from the Inner Mongolia University of Science and Technology, Baotou, China, in 2008, and the Ph.D. degree from Hunan University, Changsha, China, in 2013.

Since 2013, he has been with the School of Electrical and Information Engineering, Changsha University of Science and Technology, Changsha, where he is currently an Associate Professor. His research interests include image processing, deep learning, and sparse representation.



Yaoting Liu received the B.S. degree in automation from the University of South China, Hengyang, China, in 2020. He is currently pursuing the M.S. degree in control science and engineering with the School of Electrical and Information Engineering, Changsha University of Science and Technology, Changsha, China.

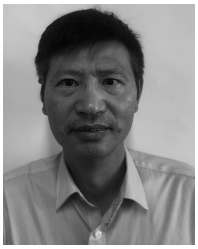
His research interests include deep learning and hyperspectral image super-resolution.



Xudong Kang (Senior Member, IEEE) received the B.S. degree from Northeast University, Shenyang, China, in 2007, and the Ph.D. degree from Hunan University, Changsha, China, in 2015.

In 2015, he joined the College of Electrical Engineering, Hunan University. His research interest includes hyperspectral feature extraction, image classification, image fusion, and anomaly detection.

Dr. Kang received the National Nature Science Award of China (Second class and Rank as third) and the Second Prize in the Student Paper Competition in IEEE International Geoscience and Remote Sensing Symposium (IGARSS) 2014. He was selected as the Best Reviewer for the IEEE GEOSCIENCE AND REMOTE SENSING LETTERS and the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING. He was an Associate Editor of the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING from 2018 to 2019. He serves as an Associate Editor for the IEEE GEOSCIENCE AND REMOTE SENSING LETTERS and the IEEE JOURNAL ON MINTATURIZATION FOR AIR AND SPACE SYSTEMS.



Shaosheng Fan received the B.S. degree from Southwest Jiaotong University, Chengdu, China, in 1987, and the Ph.D. degree from Hunan University, Changsha, China, in 2005.

Since 1996, he has been with the School of Electrical and Information Engineering, Changsha University of Science and Technology, Changsha, where he is currently a Professor. His research interests include image processing, deep learning, and electric robot.

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